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Bifunctional Phosphonium Salt-catalyzed Enantioselective [4+2] Annulation of Isoindigos with Allenes: Access to Complex Heterocycles with Centerpiece of 4*H*-Pyrans

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Abstract: Structurally fused heterocycles encompassing 4*H*-pyran centerpiece are prevalent scaffolds with promising biological activities, but have daunting synthetic challenges. Herein, we presented a regio- and stereoselective [4+2] annulation of isoindigo derivatives with allenes by bifunctional phosphonium catalysis. This protocol provides a complementary way for accessing complex heterocyclic compounds.

Keywords: complex heterocycles; 4*H*-pyran derivatives; [4+2] annulation; bifunctional phosphonium salts; H-bonding interaction

Over the past decades, asymmetric phase-transfer catalysis (PTC) has been recognized as a promising and efficient candidate for the enantioselective construction of structurally diverse and synthetically useful molecular architectures, and quaternary ammonium salt catalysis has already been extensively investigated as a powerful tool for asymmetric transformations.[1] In contrast, even though two seminal reports on phosphonium salt-catalyzed reactions can be traced back to 1990s,[2,3] the field of phosphonium salt catalysis was dormant thereafter for nearly one decade. Until recently, the key breakthroughs were communicated by Ooi[4a] and Maruoka,[5a] which refueled the field and led to the renaissance of phosphonium salt catalysis.[6] The past decade witnessed the significant advancement of asymmetric phosphonium salt catalysis, which were mainly contributed by four leading research groups of Ooi,[4] Maruoka,[5] Ma,[7] and Zhao.[8] Numerous phosphonium salt-catalyzed asymmetric transformations have been extensively studied, and the reaction scope is mainly focused on conjugate addition, Mannich reaction and substitution reaction. However, the phosphonium salt-promoted annulation reactions are rare, despite such reaction process is a powerful and versatile tool for the enantioselective synthesis of both carbocyclic and heterocyclic compounds. To the best of our knowledge, just few examples on the creation of chiral carbocyclic systems via annulation processes under amino acid-derived phosphonium salt catalysis were disclosed by Zhao and co-workers.[8j–m] Quite recently, we developed novel dipeptide-based bifunctional phosphonium salt catalysts that were successfully applied in enantioselective annulation reactions and constructing a kaleidoscope of chiral cyclic molecules.[9] Notably, such dipeptide-derived phosphonium salts possess remarkably high tunability, but their utilizations in creation of chiral cyclic systems have not been investigated in depth.

On the other hand, six-membered 4*H*-pyran heterocycles are frequently encountered as privileged structural frameworks in numerous natural products and pharmaceuticals because of their promising biological activities (Figure 1). Among many 4*H*-pyran cores, structurally fused and spirocyclic 4*H*-pyrans bearing an all-carbon quaternary stereocenter at the pyrancycle are much attractive, and a number of [4+2] annulations with commonly employing α,β-unsaturated ketones as C4-synthons were reported as efficient ways for affording such 4*H*-pyran systems.[11,12] Quite recently, our group disclosed a highly enantioselective formal [4+2] annulation between electron-deficient

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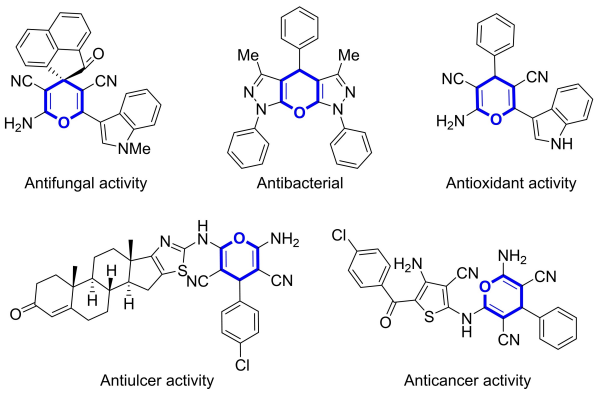
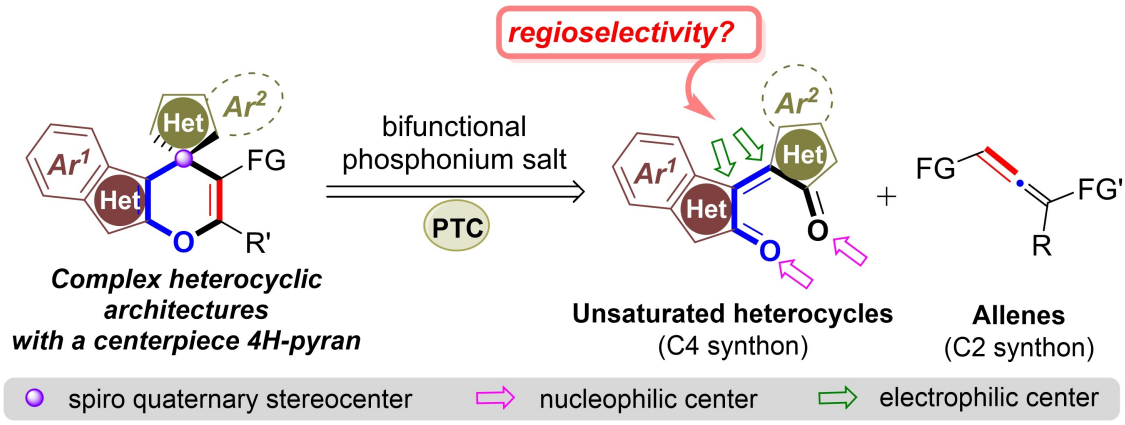


Figure 1. Selected bioactive molecules bearing 4*H*-pyran scaffold.

allenes and 3-alkenyl-oxindoles by bifunctional phosphonium salt catalysis, thus leading to facile construction of chiral heterocyclic compounds bearing 4*H*-pyran core.[9f] However, structurally complex heterocyclic architectures with a centrepiece of 4*H*-pyran still remain to be hardly available from a synthetic perspective. Virtually, such synthetic challenges faced are intrinsic: firstly, the highly crowed nature of multi-substituted alkenones makes the addition of C2-substrates extremely difficult; secondly, how to control the regioselectivity is another daunting challenge; additionally, how to regulate the reactivity of allenoates under PTC conditions[13] also needs to be solved. To expand the scope of application of this method and make the products more diverse in structure, we herein wish to update a highly enantioselective synthetic approach with employing an isoindigo as a starting substance, via phosphonium salt-catalyzed [4+2] annulation with allene, and thus lead to a variety of optically pure, structurally fused and spirocyclic 4*H*-pyran derivatives bearing a quaternary stereocenter (Scheme 1).

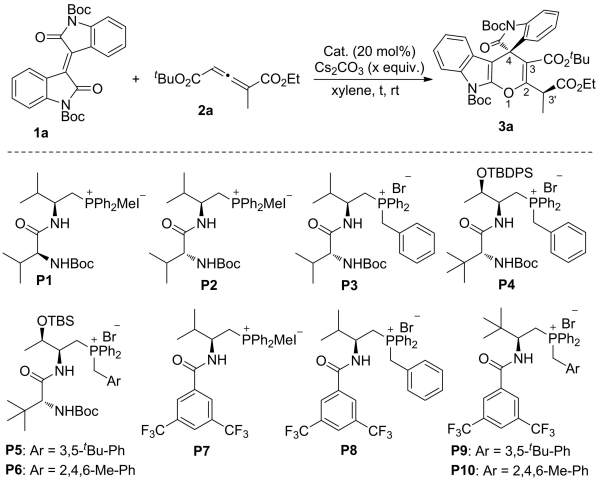
We began our investigation of [4+2] annulation between model subtrate *N*-Boc isoindigo **1a** and allenoate **2a**, in the presence of an *L*-Val-*L*-Val-based dipeptide phosphonium salt catalyst **P1** with using cesium carbonate as base. The reaction proceeded efficiently in xylene at room temperature, and the desired 4*H*-pyran product **3a** was obtained in high yield (88%) with good diastereoselectivity (97:3), but was almost racemic (Table 1, entry 1). In this context, further optimization was mainly focusing on tuning the catalyst skeletons. To our gratification, the *L*-Val-*D*-Val-based dipeptide phosphonium salt **P2** afforded the desired product **3a** in improved enantioselectivity (18% ee, entry 2). Inspired by this preliminary result, we then synthesized and examined a set of (*L, D*)-dipeptide-based chiral phosphonium salts, and found that both introduction of a bulky functionality such as *tert*-butyldiphenylsilyl or *tert*-butyldimethylsilyl to hydroxyl group and modification of alkyl halide with sterically hindered aryl ring were benefical for the stereocontrol of this reaction (entries 3–6). Among all, the O-TBS-*L*-Thr-*D*-*tert*-Leu-based phosphonium salt **P6** was recognized as the most efficient catalyst (entry 6). For comparison, various amino-acid-derived phosphonium salts with amides **P7**–**P10** were also screened under the aforementioned condition, and all these amide-based catalyst gave the cyclized product **3a** in low to moderate ee values (entries 7–10). Subsequently, the effects of solvent and base additives were routinely conducted, and no better results were obtained (Table S2 and Table S3).[14a] Additionally, when 4.0 equivalents of base Cs_2CO_3 was applied, the reaction time was shortened and the enantioselectivity was slightly enhanced (entry 12). As expected, decreasing the reaction temperature to −20 °C could improve the enantioselectivity for giving 96% *ee*



Scheme 1. Construction of complex heterocycles with a centrepiece 4*H*-pyran via bifunctional phosphonium salt-catalyzed [4+2] strategy.

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Table 1. Optimization of the reaction conditions[a]



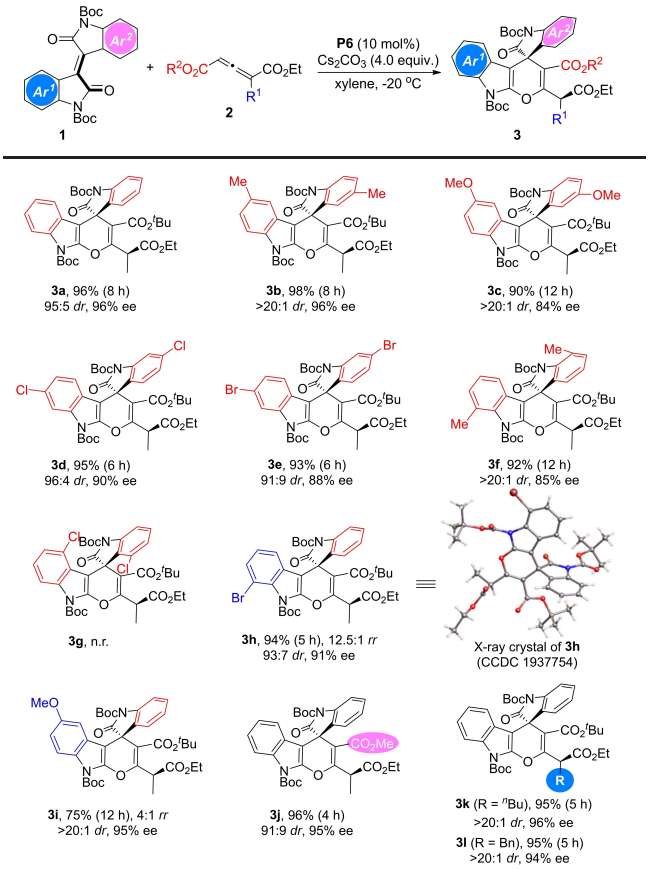
Entry	Cat.	× (equiv.), t	Yield (%)[b]	dr	ee (%)[c]
1	**P1**	2, 1 h	88	97:3	2
2	**P2**	2, 2 h	91	94:6	18
3	**P3**	2, 1 h	85	94:6	12
4	**P4**	2, 8 h	87	94:6	45
5	**P5**	2, 1 h	90	96:4	75
6	**P6**	2, 0.5 h	93	95:5	88
7	**P7**	2, 1 h	89	96:4	8
8	**P8**	2, 7 h	87	95:5	26
9	**P9**	2, 20 h	77	96:4	44
10	**P10**	2, 5 h	84	90:10	69
11	**P6**	1, 1 h	95	95:5	88
12	**P6**	4, 0.3 h	95	94:6	90
13[d]	**P6**	4, 2 h	94	94:6	94
14[e]	**P6**	4, 6 h	96	96:4	96
15[e,f]	**P6**	4, 8 h	96	95:5	96

[a] Reactions were performed with **1a** (0.05 equiv.), **2a** (0.075 equiv.), catalyst (20 mol%), the Cs_2CO_3 (× equiv.) in xylene (1.0 mL) at room temperature. All *dr* values were determined by ^{1}H NMR of crude product. TBS = *tert*-butyldimethylsilyl. OTBDPS = *tert*-butyldiphenylsilyl.
[b] Isolated yield.
[c] The *ee* values were determined by HPLC analysis.
[d] At 0 °C.
[e] At −20 °C.
[f] The catalyst loading was 10 mol%.

(entry 14). It is worth noting that the reactivity and stereoselectivity were maintained even though a 10 mol% catalyst loading was employed (entry 15).

With the current optimized condition in hand (Table 1, entry 15), the substrate scope of isoindigos was investigated (Table 2). It was found that isoindigos with *N*-Boc group bearing both electron-donating and electron-withdrawing substituent groups on oxindole moiety were well tolerated for this reaction, providing relevant products **3a–f** in excellent yields, diastereo- and enantioselectivities (up to 98% yield, >20:1 *dr*

Table 2. Scope for [4+2] annulation between isoindigos **1** and allenes **2** [a,b,c]

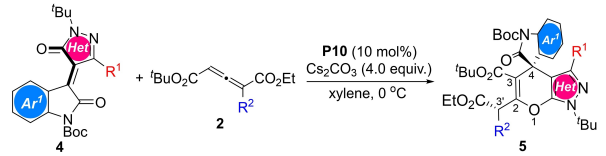


[a] Reactions were performed with **1** (0.10 mmol), **2** (0.15 mmol), catalyst **P6** (10 mol%) and Cs_2CO_3 (0.40 mmol) in xylene (2.0 mL) at −20 °C for 4–12 h.
[b] Isolated yields.
[c] All *dr* and *rr* values were determined by ^{1}H NMR or HPLC analysis of crude products, and the *ee* values were determined by HPLC analysis.

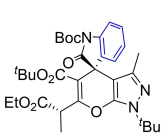
and up to 96% *ee*). When isoindigo **1g** with chloride group at C4-position was used, the annulation product **3g** was not observed, probably due to the crowded circumstance around the all-carbon quaternary center. Unsymmetrical isoindigos were then evaluated. In the perspective of synthetic methodology, isoindigos containing oxindoles with different substituents are more challenging, that usually lead to intricate regioisomeric products if the reaction is with a poor regioselectivity.[15] Fortunately, the [4+2] annulation of unsymmetrical isoindigos was precisely controlled by our catalytic system, giving annulation products **3h** and **3i** in regioselective and stereoselective manners. Later, the substitution effects of allenoates **2** were finally examined. The replacement of *tert*-butyl ester or methyl group for allenoates with other substituents had little influence on the reactivity and stereoselectivity, furnishing the products **3j–l** in 95–96%

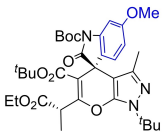
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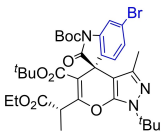
Table 3. Asymmetric [4+2] annulation between pyrazoloneyldiene oxindoles **4** and allenes **2** [a,b,c]

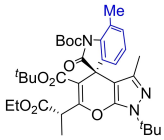


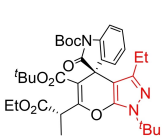
4 2 5

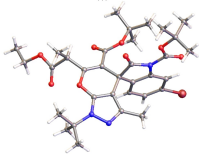

5a, 96% (3.5 h), 19:1 *rr*
94:6 *dr*, 92% ee

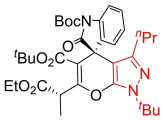

5b, 92% (5 h), >20:1 *rr*
96:4 *dr*, 90% ee

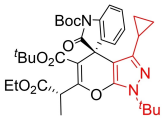

5c, 93% (6 h), 19:1 *rr*
94:6 *dr*, 97% ee[d]

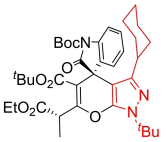

5d, 92% (9 h), >20:1 *rr*
97:3 *dr*, 92% ee

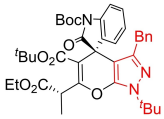

5e, 90% (6 h), 16:1 *rr*
93:7 *dr*, 93% ee

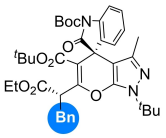

X-ray crystal of **5c**
(CCDC 1968335)

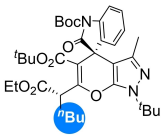

5f, 86% (6 h), >20:1 *rr*
94:6 *dr*, 87% ee


5g, 88% (6 h), 5.3:1 *rr*
94:6 *dr*, 93% ee


5h, 82% (6 h), >20:1 *rr*
97:3 *dr*, 90% ee


5i, 83% (6 h), 7:1 *rr*
94:6 *dr*, 82% ee


5j, 85% (24 h), >20:1 *rr*
98:2 *dr*, 90% ee


5k, 88% (9 h), >20:1 *rr*
97:3 *dr*, 88% ee

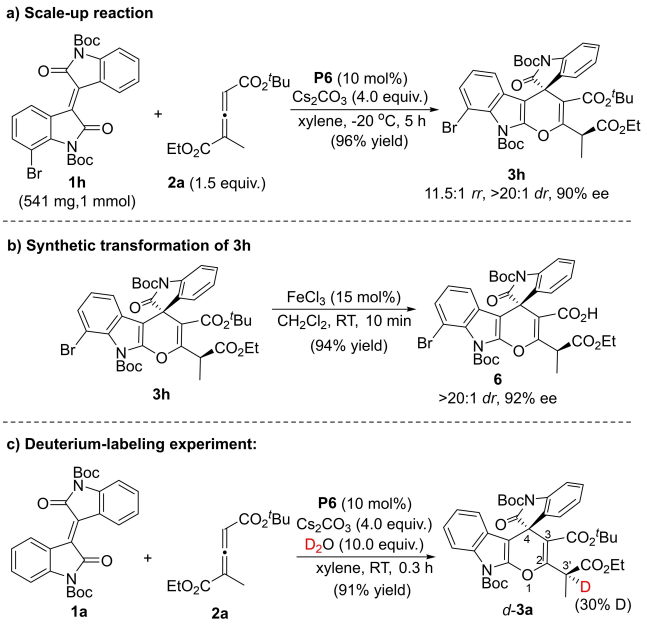
[a] Reactions were performed with **4** (0.10 mmol), **2** (0.15 mmol), catalyst **P10** (10 mol%) and Cs_2CO_3 (0.40 mmol) in xylene (2.0 mL) at 0 °C for 3.5–24 h.
[b] Isolated yields.
[c] All *dr* and *rr* values were determined by ^{1}H NMR or HPLC analysis of crude products, and the *ee* values were determined by HPLC analysis.

yield with 91:9 to >20:1 *dr* and 94–96% *ee*. Moreover, absolute configuration of **3h** was determined as (3'*S*, 4*S*) by the X-ray analysis (see SI for details).[16]

Whereafter, we turned our interest to the annulation of allenoates **2** and pyrazoloneyldiene oxindoles **4** to synthesize pyrano[2,3-c]-pyrazole derivatives **5**, which are structural analogues of potential drug candidates.[17] To our delight, the reaction could also proceed smoothly when we investigated the generality of pyrazoloneyldiene oxindoles **4** with allenoates **2** after a carefully systematic rescreening of catalyst.[14b] Of note, high reactivities with good regioselectivities and stereoselectivities were achieved under the reoptimized condition (entry 3 in Table S9). In the presence of 10 mol% chiral phosphonium salt **P10** and 4.0 equivalent Cs_2CO_3, substrates **4** with different substituents on the oxindole and pyrazolone moieties were investigated, furnishing the desired products **5a–i** in high yields with good regio-, diastereo- and enantioselectivities (up to 96% yield, up to >20:1 *rr*, 97:3 *dr* and 97% *ee*). In addition, allenoates with benzyl and *n*-butyl group were also allowed to react with pyrazoloneyldiene oxindole **4a** to afford regioselective and stereoselective annulation products **5j** and **5k** respectively. Finally, the relative and absolute configuration of **5c** was determined as (3'*S*, 4*S*) by the X-ray analysis (see SI for details).[16]

In order to prove the effectiveness of this [4+2] annulation reaction, we carried out the preparative scaled synthesis of **3h** (Scheme 2a). It was gratifying that both reactivity and stereoselectivities were maintained, for providing the desired product in 96% yield with 92:8 *dr* and 90% *ee*. In addition, treatment of **3h** with ferric chloride led to a selective hydrolysis process and furnished the hydrolysed compound **6** in good isolated yield with coincident stereoselectivities (Scheme 2b).

Based on these results and our previous studies,[9a–b] a plausible reaction cycle was presented in Figure S3.[14c] To demonstrate our hypothesis, the deuterium-labelling experiment was further conducted to get preliminary supporting for this catalytic cycle. As shown in Scheme 2c, when the model reaction was performed in the presence of 10.0 equivalents of additive D_2O, deuterium incorporation into 3'-position (30% D) of the cyclized product was observed, which

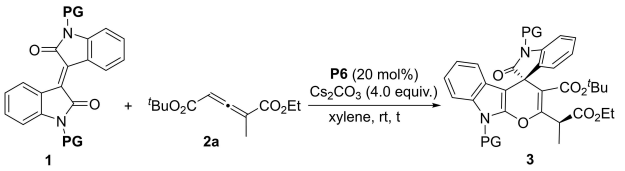


Scheme 2. Scale-up reaction, synthetic transformation and deuterium labelling experiment.

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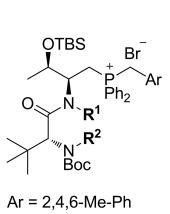
indicated the beneficial existence of possible intermediate **D**/**E** in this reaction system (Figure S3). Moreover, to gain some insights on vital stereo-control mode of this transformation, we further prepared the methylated catalyst **P6–1** and **P6–2** to view the significance of H-bonding interactions for stereo-induction. When either of methylated catalyst was used for the model reaction, both diastereo- and enantio-selectivities were decreased substantially (Table 4, entries 1–3). Of note, when *N*-Methyl isoindigo was employed, the corresponding product was isolated in good yield but with a racemic version (entry 4), and we guess that the *N*-Boc-protected isoindigo substrate may be crucial for stereo-selectivity of the reaction. However, when the model reaction was performed in methanol, messy reaction was observed without formation of any major products (entry 5), and we suspect this may due to the instability of allene substrate under slightly stronger basic system induced by polar solvent. These experimental results indicated that the weak-bond interactions including H-bonding and ion-pair manners were critical for the excellent stereoselectivities under this bifunctional phosphonium catalysis, and thus a plausible transition state model (**TS1**) was proposed with respect to the key step of intermolecular α-addition between allene **2 a** and isoindigo **1 a**.

Table 4. Control experiments[a]



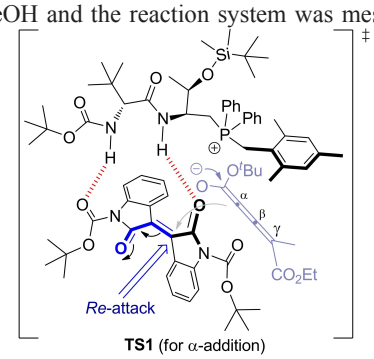
Entry	Cat.	PG	t (h)	Yield (%)	dr/ee (%)
1	**P6**	Boc	0.3	95	95:5/90
2	**P6–1**	Boc	2	90	84:16/25
3	**P6–2**	Boc	5	84	71:29/86
4	**P6**	Me	5	94	90:10/0
5[e]	**P6**	Boc	1 min	–	–

[a] Reactions were performed with **1** (0.05 mmol), **2 a** (0.075 mmol), Cs_2CO_3 (0.20 mmol) and catalyst (0.01 mmol) in xylene (1.0 mL) at room temperature.
[b] Isolated yield.
[c] The *dr* values were determined by ^{1}H NMR of crude product.
[d] The *ee* values were determined by HPLC analysis.
[e] The solvent was MeOH and the reaction system was messy.



Ar = 2,4,6-Me-Ph

P6: $R^1 = R^2 = H$
P6-1: R^1 = Me, R^2 = H
P6-2: R^1 = H, R^2 = Me



In conclusion, we disclosed a highly efficient phosphonium salt-catalyzed enantioselective [4+2] annulation reaction between allenes and multi-substituted alkenones containing two identical/different unsaturated heterocyclic moieties for the construction of structurally fused and spirocyclic 4*H*-pyran core frameworks. By employing an amino acid-derived bifunctional phosphonium salt as phase-transfer catalyst, a wide range of synthetically challenging, multiply substituted and structurally fused heterocycles with spiro-4*H*-pyran units were conveniently constructed in high yields with excellent diastereo- and enantioselectivities under mild conditions. In addition, this annulation process is featured with very good regioselectivities, and is highly useful for creating complicated and challenging heterocyclic compounds. The practicality and the utility of this protocol were demonstrated by the scale-up synthetic reaction and facile elaboration. Currently, further developments of challenging asymmetric processes and their broadly synthetic applications with such bifunctional phosphonium salt catalysis are ongoing in our laboratory.

Experimental Section

General Procedure for Asymmetric Synthesis of 3: A mixture of isoindigos **1** (0.1 mmol), Cs_2CO_3 (0.4 mmol), the catalyst **P6** (0.01 mmol) and the allenoates **2** (0.15 mmol) in xylene (2.0 mL) was stirred at −20 °C until that the reaction was completed (monitored by TLC). Then the reaction mixture was filtered to remove residual base, and the solid powder was washed with ethyl acetate (2.0 mL). After evaporation under the reduced pressure, the residue was purified through flash column chromatography on silica gel (petroleum ether/ethyl acetate = 8:1 to 5:1) to afford the products **3**.

General Procedure for Asymmetric Synthesis of 5: A mixture of pyrazoloneyldiene oxindoles **4** (0.1 mmol), Cs_2CO_3 (0.4 mmol), the catalyst **P10** (0.01 mmol) and the allenoates **2** (0.15 mmol) in xylene (2.0 mL) was stirred at 0 °C until the reaction was completed (monitored by TLC). Then the reaction mixture was filtered to remove residual base, and the solid powder was washed with ethyl acetate (2.0 mL). After evaporation under the reduced pressure, the residue was purified through flash column chromatography on silica gel (petroleum ether/ethyl acetate = 8:1 to 5:1) to afford the desired products **5**.

Acknowledgements

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